

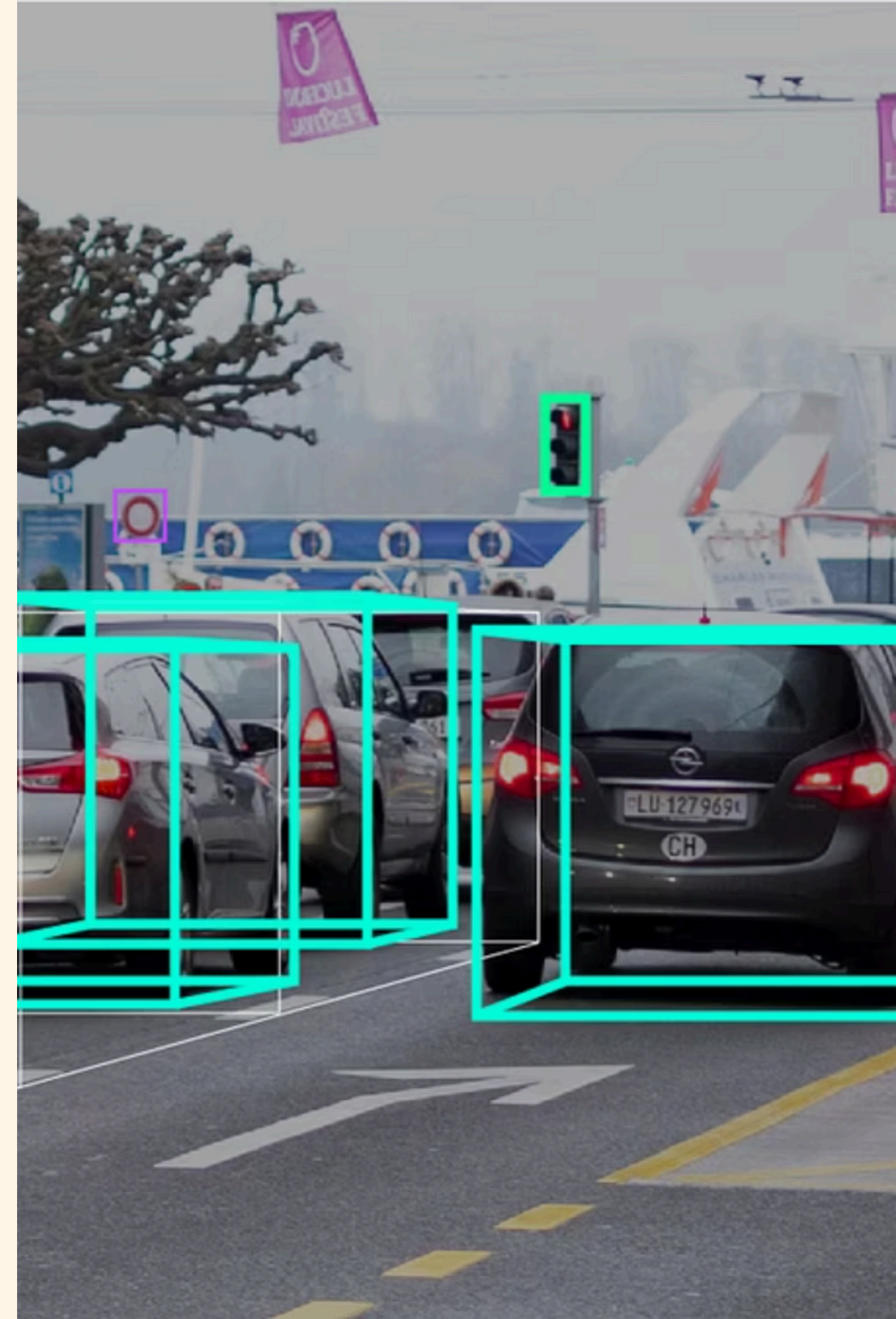
Object Detection using YOLOv4

Object detection identifies **what objects are present and where they are located** in an image

Unlike image classification, it returns **bounding boxes and class labels**

This project demonstrates **real-world object detection** using a deep learning model

Goal: Detect multiple objects from a user-uploaded image efficiently



YOLOv4, COCO Dataset & Detection Logic

YOLO (You Only Look Once)

- Single-stage object detection algorithm
- Processes the entire image **in one forward pass**
- Predicts bounding boxes, objectness score, and class probabilities
- Optimized for **speed and real-time performance**

YOLOv4

- Improved version of YOLO with better accuracy
- Uses CSPDarknet53 backbone
- Effective for detecting small and large objects

YOLOv4, COCO Dataset & Detection Logic

COCO Dataset

- Large-scale dataset with **80 common object classes**
- Includes people, vehicles, animals, and daily objects
- Pre-trained weights allow detection **without training the model**

Detection Thresholds

- **Objectness Threshold:** Filters weak object proposals
- **Confidence Threshold:** Ensures reliable class predictions
- **Non-Maximum Suppression (NMS):** Removes duplicate overlapping boxes

Methodology and Key Learnings

- User uploads an image
- Image is resized and normalized before detection
- YOLOv4 performs object detection
- Confidence filtering and NMS refine results
- Bounding boxes and labels are drawn on detected objects

Key Learnings

- Difference between classification and object detection
- Practical use of pre-trained deep learning models
- Importance of post-processing techniques like NMS
- Limitation: detects only COCO-trained object categories